

Ian Merriman's Drogue-Chute Nose-Cone Project:

I created a modified nose cone for my L1 certification rocket which self-deploys after the ejection charge fires at the end of the main motor burn.

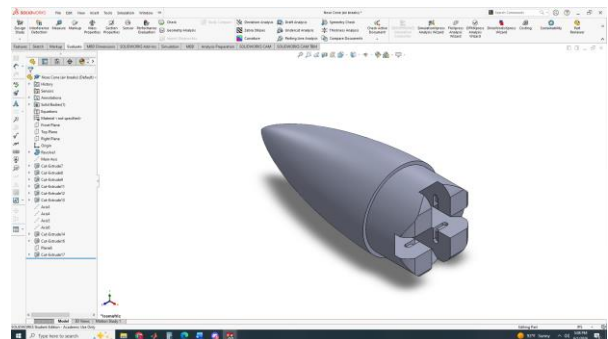
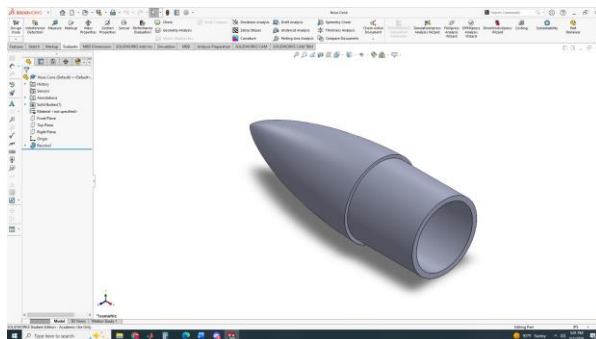
The airbrakes splay out, increasing the drag and ensuring that the parachute is successfully drawn out of the body tube. This system was developed after my first rocket crashed due to the parachute not deploying properly.

First Launch Result:

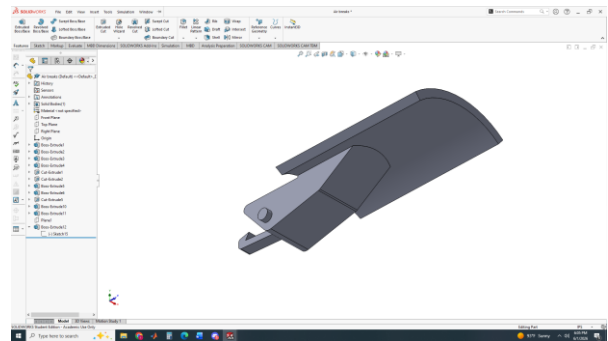
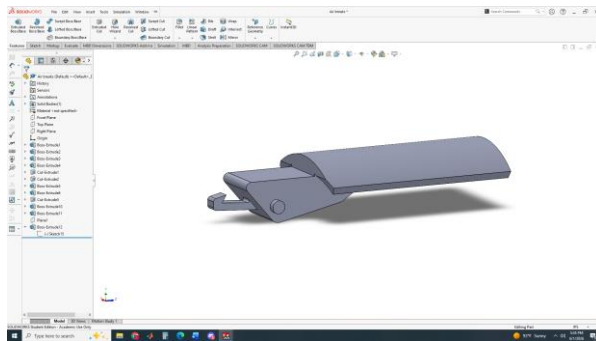


Nose-Cone Redesign:

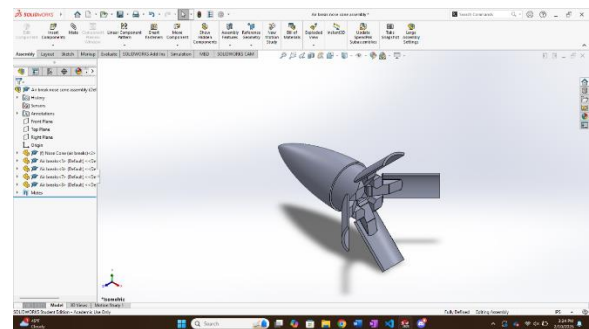
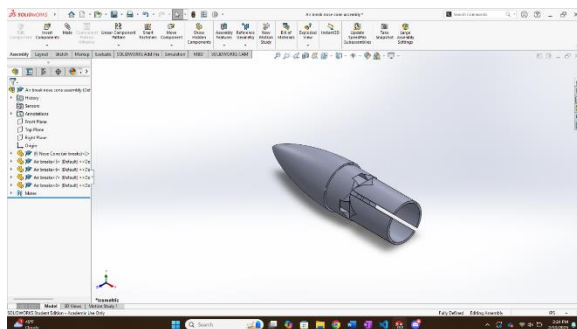
I started by taking my original nose-cone design, shortening the shoulder section, making the shoulder walls thicker. From there, I removed material in an X-pattern to create room for the arms of the airbrakes and added slots for their pivot bars.



For the airbrakes, I gave the fin sections the same curvature as the original nose-cone's shoulder but split them into 4 sections. Since the brakes would be made from weak PLA, the body-section was made thick and durable to withstand the expected impact and drag forces. Originally, the stubs on the side were meant to be the pivot points, made from the same material, however, they were replaced with steel rods, cut to size via dremel, which ran through the pieces to avoid the possibility of breakage under loading. Finally, a catch arm was added so that a rubber-band could be stretched between all 4 airbrakes, making them ready to flare out once removed from the rocket's body tube.



SolidWorks Assembly:



After-Print Modifications:

Nails were added as columns to further stretch the rubber band and make the action quicker as well as keep the pivot rods locked in place. The nose-cone's airbrakes open as soon as it leaves the tube, due to the rubber-band.



To increase drag further, fabric was epoxied onto the airbrakes to fill the gaps when fully extended.



Testing:

The modified nose-cone was tested on it's effectiveness prior to flight and resulted in a 100% success rate of dislodging parachutes at a windspeed of 35 mph, which the rocket would easily reach if the parachute remained lodged in the body.

Result:

Successfully deployed during test flight resulting in L1 certification through Tripoli Rocketry Association.



Attempt for Level <u>1</u> High-Power Certification TRA# <u>27056</u>				
Flyer's name (Print) <u>Jon Merriam</u>				
Rocket information: If an item is not applicable write NA				
Kit name/manufacturer (or scratchbuilt)	Length/Diameter (in.)	Weight (lb.)	CP/CG (in. from nose tip)	Altitude expected (ft)
<u>Scratchbuilt</u>	<u>38 / 3.55</u>	<u>2.7</u>	<u>20.5 / 29.7</u>	<u>1850</u>
Electronics	<u>none</u>			
Recovery	<u>electronic recovery, motor ejection charge</u>			
Motor (manufacturer, designation)	<u>H100W-14A</u>			
Other comments / Mentor TRA #, Name & Signature for TMP L1 (use M1 to indicate in Attempt for Level)				
Check each applicable statement (Some assistance from a certified flyer in each case is ok): ☒ I built this rocket myself (either from a kit or from parts). ☒ I prepared this rocket for flight myself. ☒ I assembled the motor myself. ☒ (For L3 only) I have flown a J/N/L rocket rocket with successful electronic deployment for recovery.				
Flyer's signature <u>Jon Merriam</u>		Date <u>5/25/24</u>		
Following is for Certifying Authority Use Only. Certifying Authority shall send this form to Tripoli Headquarters following the certification flight attempt. May only be signed by a Prefect, TAP or Board Member.				
For L2 Certification: Written Examination				
Examination Score		Examination Prefect Signature/Date		
For L3 Certification: TAP Approval				
Design and documentation of this Level 3 project has been reviewed and found satisfactory.				
TAP signature/TRA/Date		TAP signature/TRA/Date		
For All Certifications: Flight result ☒ Successful ☐ Unsuccessful				
Launch location/venue <u>SG-1402.0 R. TX</u>				
Comments— if flight was unsuccessful, give the reason(s)				
Certifying Authority Signature/TRA# <u>SG-1402.0 R. TX</u> <u>14375</u>				
Certifying Authority: Regardless of flight outcome, send this form to: Tripoli Rocketry Association, Inc. P. O. Box 87 Bellevue, NE 68005				